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Roll No

MI-8001 (CBGS)
B.E. VIII Semester Examination, June 2020
Choice Based Grading System (CBGS)
Mining Machinery - III
Time : Three Hours

Maximum Marks : 70

- Note:** i) Attempt any five questions.
ii) All questions carry equal marks.
iii) Assume suitable data wherever necessary.

1. a) Describe the main features of Jumbo Drill machine and its various types.
b) What are various types of coal cutting machine? How they differ in their cutting operation and working? Explain.
2. a) Describe the constructional features of a Double Ended Ranging Drum Shearer. Explain the various auxiliary loading devices.
b) Describe the main constructional features of a Road Heading Machine with a neat sketch.
3. Describe the constructional features of any two
 - i) LHD
 - ii) Drill Jumbo
 - iii) U/G Dump trucks
4. a) Draw a neat layout of a longwall face equipped with Coal Plow. What are the various types of Plow?
b) What is Sumping? Explain full face triangular inclined sumping step by step with neat diagram.
5. Give major classifications of compressors and the basic principle of its working.
6. a) Write notes on
 - i) Effect of water on compressor performance
 - ii) Safety features in compressors
b) List the components of a compressed air system and explain their functions.
7. a) What is Gate End Box? What are the various safety and protection devices provided in the Gate End Box?
b) Write notes:
 - i) Intrinsically safe circuit
 - ii) Trailing cables and their construction
8. a) Discuss the design features of FLP enclosure for use in Underground coal mines.
b) Compare applicability and performance of Shearer with Coal Plow.

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